

Build a Personalized Online Course Recommender System with Machine Learning

Content-based and collaborative-filtering recommenders for AI
Training Room

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Recommended for you

 Python

Python for Data Science

 Machine Learning

**Machine Learning with
Python**

 Big Data

Big Data 101

 Data Science

**Data Science
Methodology**

 Machine Learning

Deep Learning 101

 Cloud Computing

Introduction to Cloud

Outline

- Introduction and Background
- Exploratory Data Analysis
- Content-based Recommender System using Unsupervised Learning
- Collaborative-filtering based Recommender System using Supervised learning
- Conclusion
- Appendix

Introduction

Project background and context

AI Training Room is an online learning platform serving a large learner base across a catalogue of 307 courses. With 33,901 learners and more than 233,000 recorded course interactions, manually finding relevant courses becomes increasingly difficult. A recommender system can help learners discover useful courses faster, build more personalized learning paths, and potentially improve engagement and course enrollment.

Problem statement

How can AI Training Room use learners' previous course activity and course characteristics to recommend relevant new courses that each learner has not already taken?

307
courses

33,901
learners

233,306
enrollments

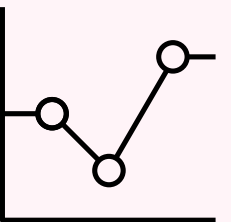
Hypotheses

H1. Learners' previous course genres can be used to identify other courses that match their interests.

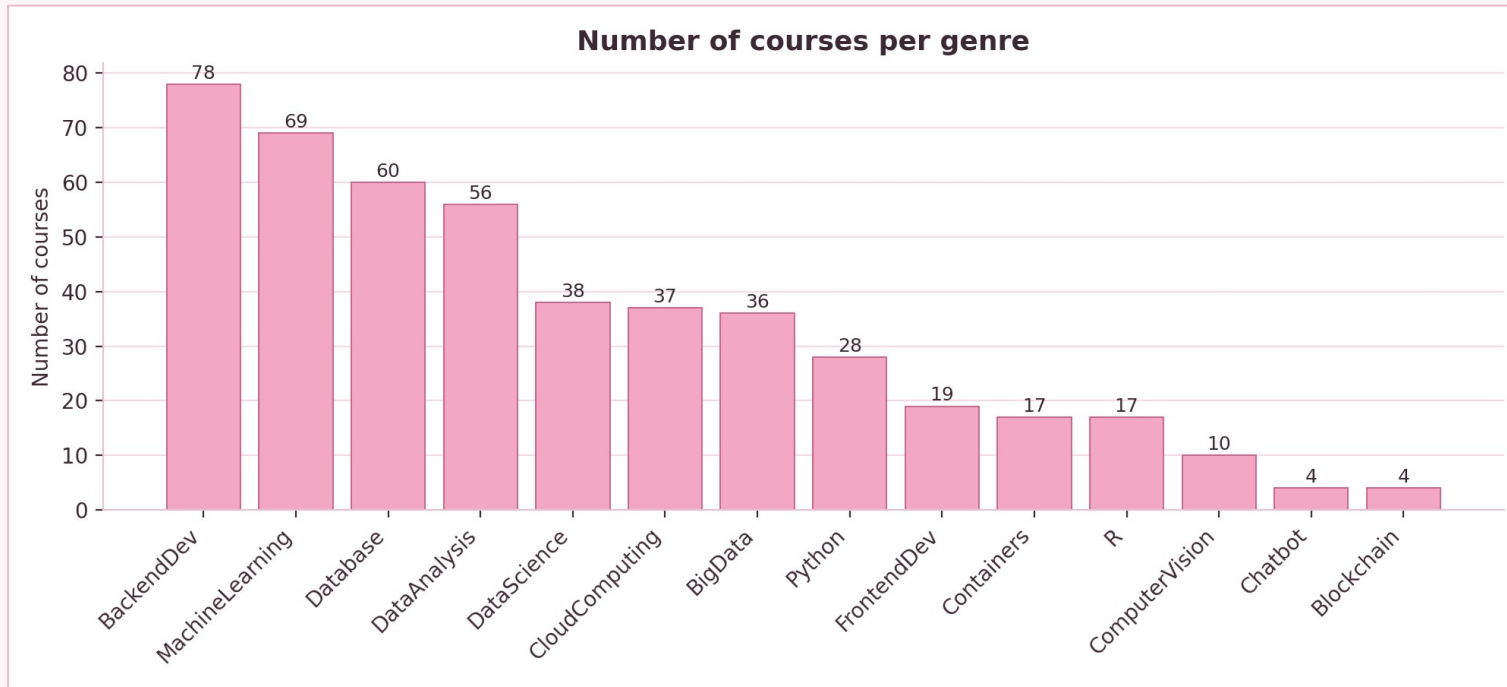
H2. Collaborative filtering methods will predict learner ratings more accurately than simply predicting the average rating.

H3. Neural-network embeddings will outperform KNN and NMF by learning richer relationships between learners and courses.

Exploratory Data Analysis

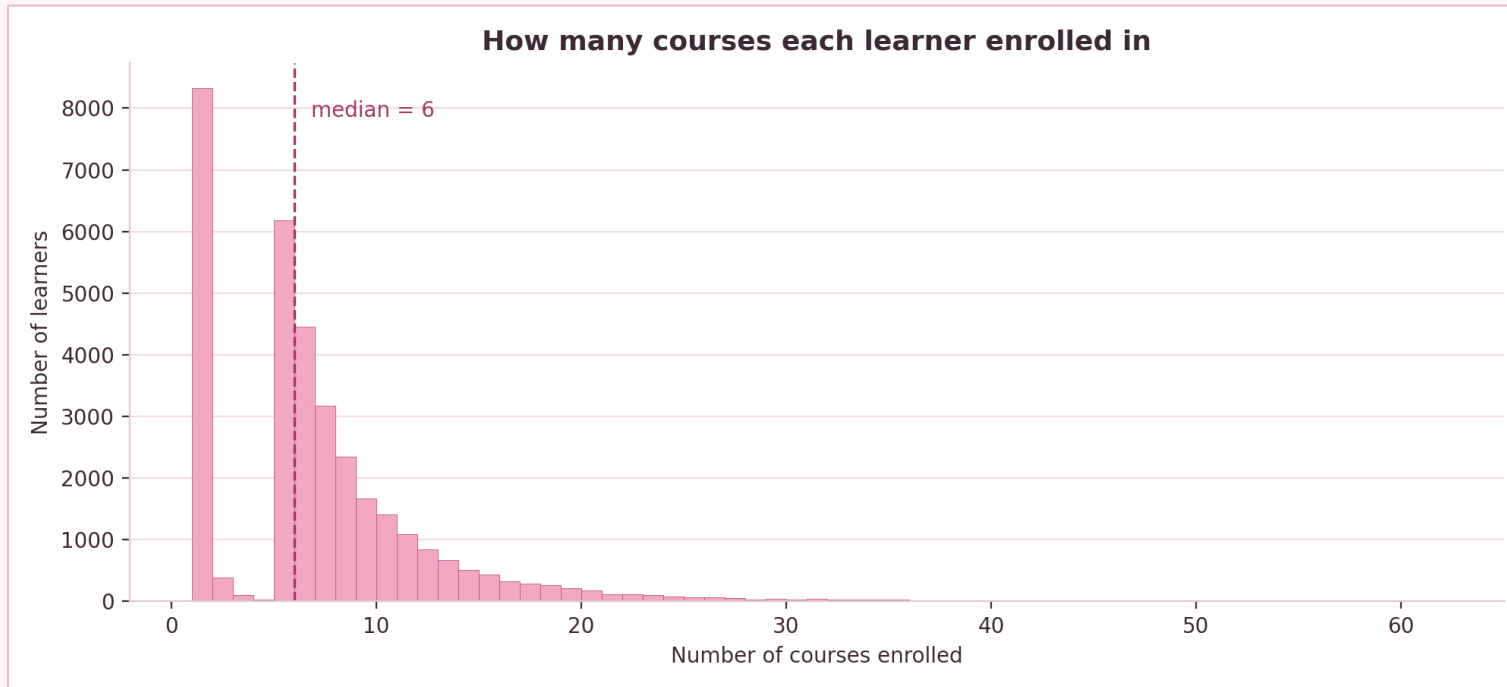


Course counts per genre



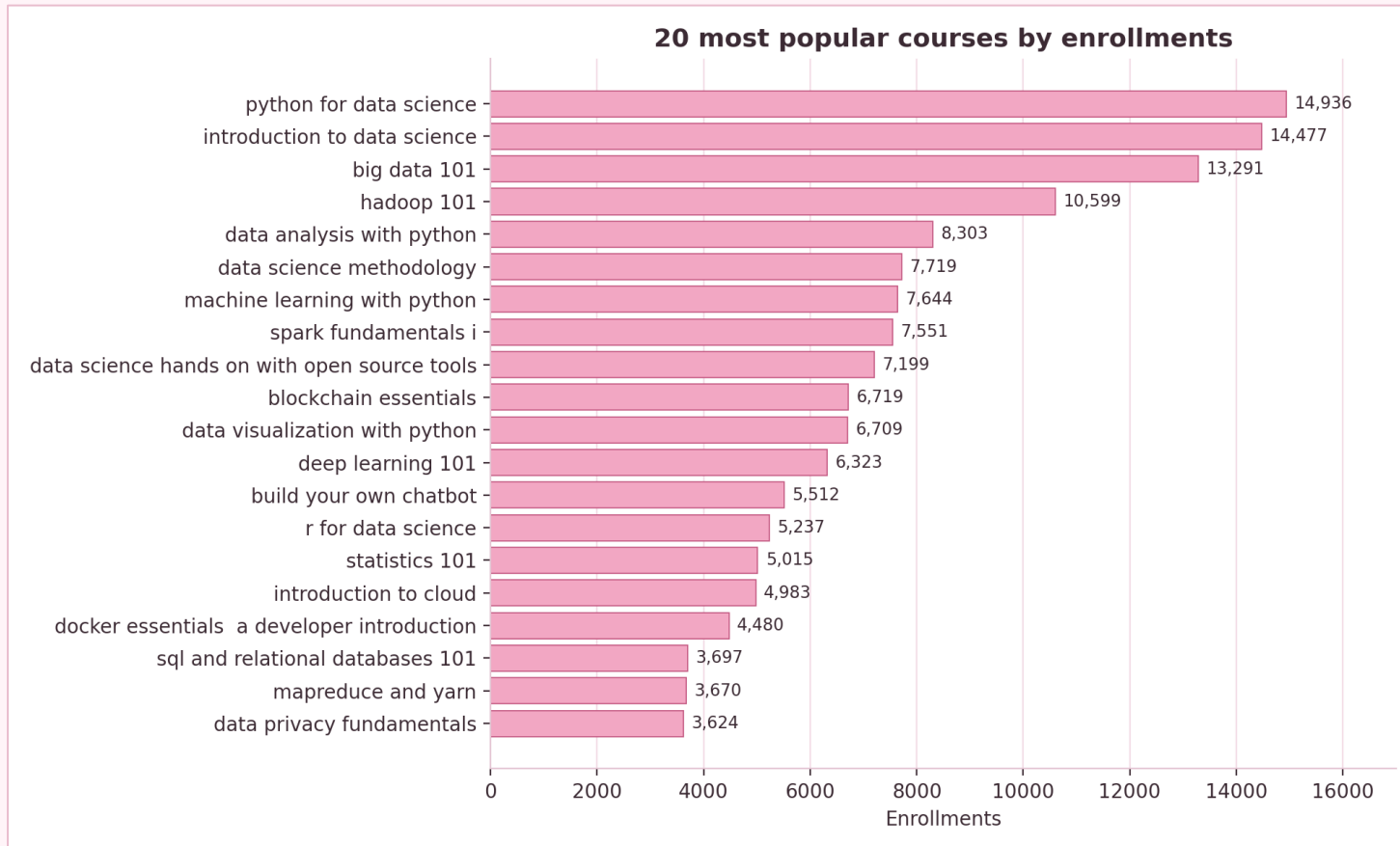
- BackendDev (78), MachineLearning (69) and Database (60) are the largest genres.
- Chatbot and Blockchain have only 4 courses each, so learners interested in them have few options.
- A course can belong to several genres, so the counts add up to more than the 307 courses.
- The genre vectors (0/1 per genre) are the course features used by the profile-based recommender.

Course enrollment distribution



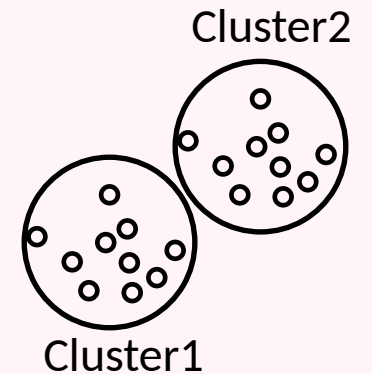
- Enrollments per learner are strongly right-skewed: median 6 courses, mean 6.9, maximum 61.
- 8,320 learners (about 25%) enrolled in only one course.
- Implication: many users have very little history (cold start), which limits collaborative filtering and makes content-based methods valuable.

20 most popular courses



- Python for Data Science (14,936), Introduction to Data Science (14,477) and Big Data 101 (13,291) are the most popular courses.
- The top 20 of 126 enrolled courses account for 63.3% of all enrollments.
- Risk: popularity bias. A naive recommender would keep pushing the same introductory courses and rarely surface niche ones.

Content-based Recommender System using Unsupervised Learning



Flowchart of content-based recommender system using user profile and course genres

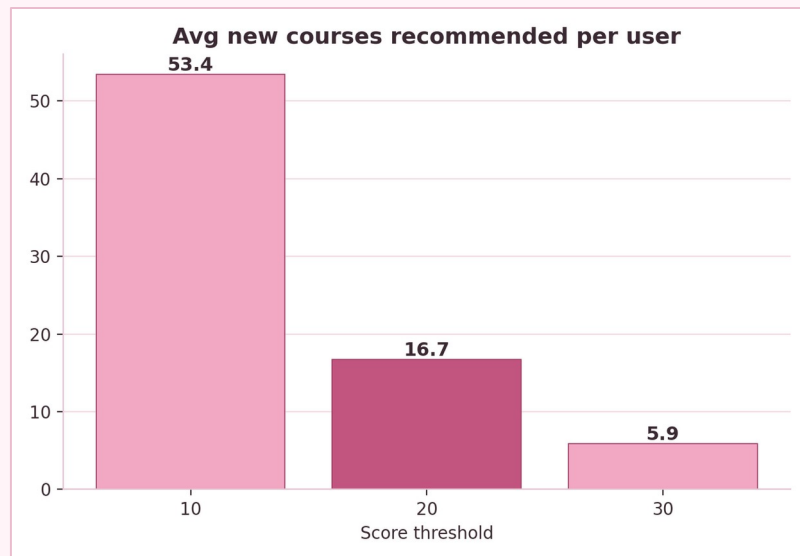


How it works: a learner's interest profile is built from the genres of courses they already took. Each unseen course is scored by how well its genres match that profile (dot product), and courses above the score threshold are recommended. Tested thresholds: 10, 20 and 30.

Evaluation results of user profile-based recommender system

Hyper-parameters: score threshold on the profile-course dot product, tested at 10, 20 and 30; top-10 list shown for threshold = 10.

On average, how many new/unseen courses were recommended per user?



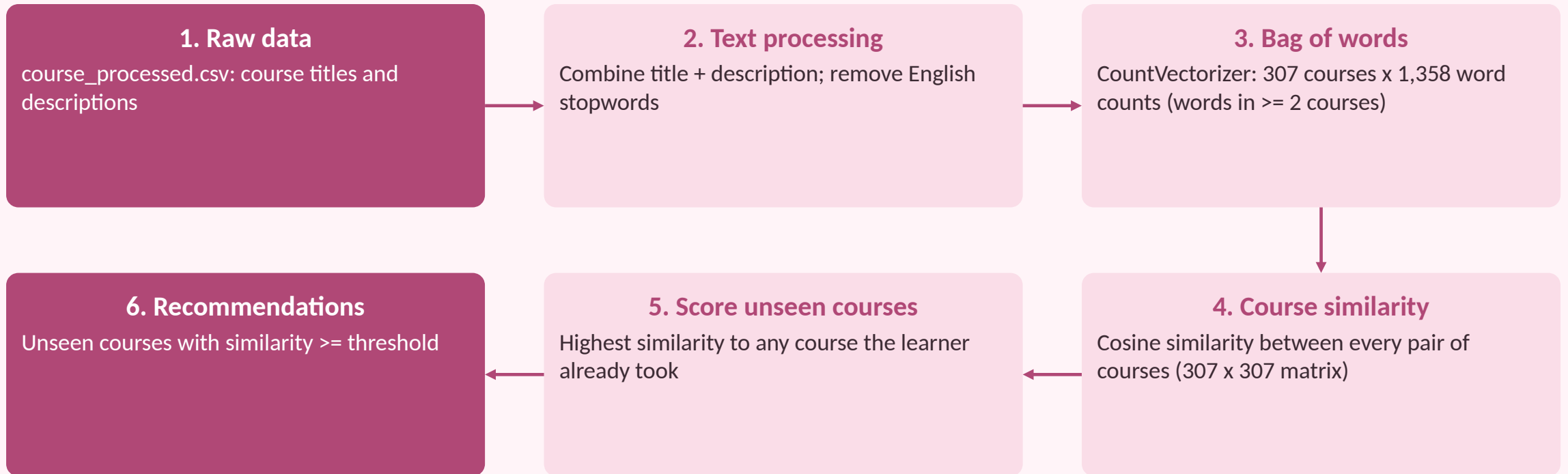
53.4 new courses per user at threshold 10, 16.7 at 20 and 5.9 at 30.

Threshold 20 (about 17 courses) gives a practical balance between coverage and precision. Courses tagged with many genres score high for most users, so broad multi-genre courses dominate the top-10.

Top-10 most frequently recommended courses

#	Course ID	Title	Users
1	TA0106EN	text analytics at scale	608
2	GPXX0IBEN	data science in insurance: basic statistical analysis	548
3	excourse22	introduction to data science in python	547
4	excourse21	applied machine learning in python	547
5	ML0122EN	accelerating deep learning with gpu	544
6	excourse06	sql for data science capstone project	533
7	GPXX0TY1EN	performing database operations in cloudant	533
8	excourse04	sql for data science	533
9	excourse31	cloud computing applications part 2: big data	524
10	excourse72	foundations for big data analysis with sql	516

Flowchart of content-based recommender system using course similarity

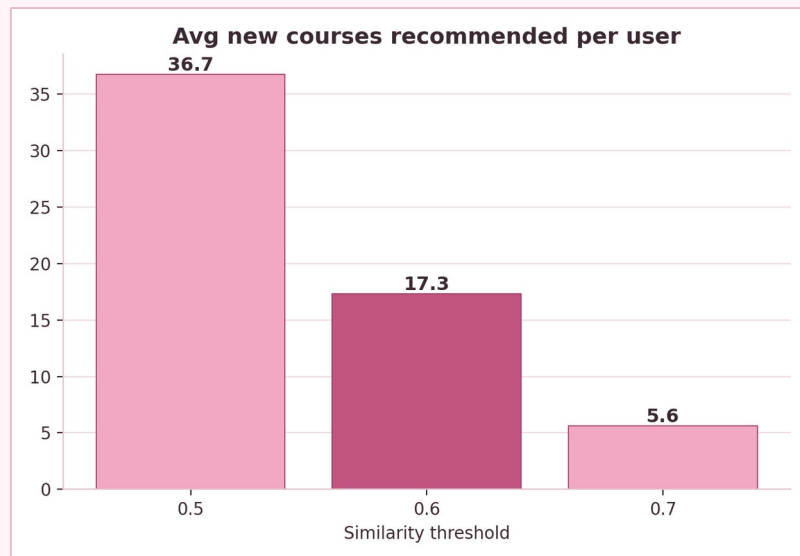


How it works: each course is represented by the words in its title and description. Courses that share many words are similar, so a learner who took course X is recommended unseen courses whose text is most similar to X ("because you took X"). Tested thresholds: 0.5, 0.6 and 0.7.

Evaluation results of course similarity based recommender system

Hyper-parameters: cosine similarity threshold, tested at 0.5, 0.6 and 0.7; top-10 list shown for threshold = 0.6.

On average, how many new/unseen courses were recommended per user?



36.7 new courses per user at 0.5, 17.3 at 0.6 and 5.6 at 0.7.

Recommendations focus on data science, data analysis and Python. Near-duplicate catalogue entries (e.g. "data analysis using python" appears twice) show that very similar courses should be de-duplicated or filtered out.

Top-10 most frequently recommended courses

#	Course ID	Title	Users
1	excourse68	big data modeling and management systems	721
2	excourse32	introduction to data analytics	718
3	excourse04	sql for data science	677
4	excourse23	data analysis using python	677
5	excourse36	data analysis using python	677
6	excourse64	data science in real life	672
7	DS0110EN	data science with open data	600
8	excourse22	introduction to data science in python	579
9	excourse62	introduction to data science in python	579
10	TMP107	data science bootcamp with python	555

Flowchart of clustering-based recommender system

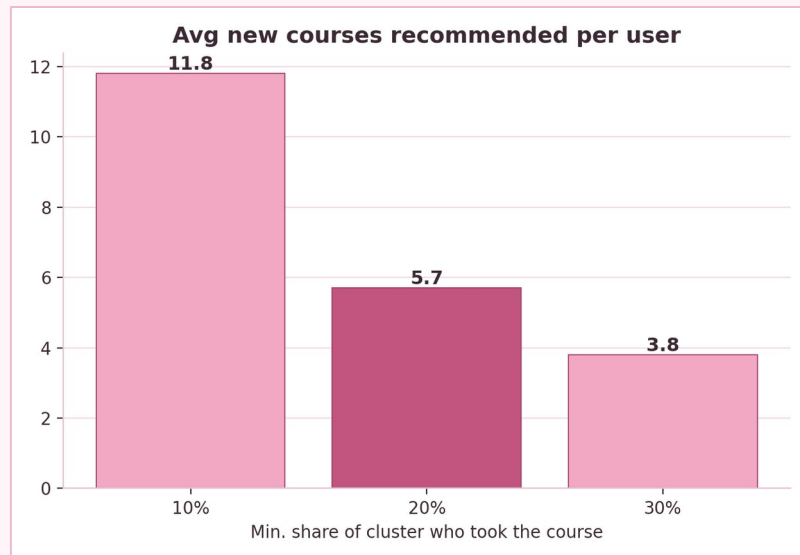


How it works: learners are grouped by their genre interests (after PCA dimensionality reduction). Each learner is recommended courses that are popular among learners in the same cluster but that they have not taken yet ("learners like you also took"). Tested shares: 10%, 20% and 30%.

Evaluation results of clustering-based recommender system

Hyper-parameters: PCA with 9 components (90% variance), K-Means with $k = 15$, minimum share of cluster tested at 10%, 20% and 30%; top-10 for 20%.

On average, how many new/unseen courses were recommended per user?



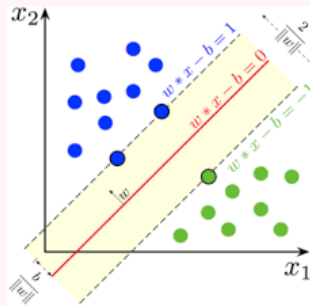
11.8 new courses per user at 10%, 5.7 at 20% and 3.8 at 30%.

The most conservative method: it recommends mainstream foundation courses (all in the overall top-20), so it has the strongest popularity bias. Cluster sizes range from 20 to 8,629 learners.

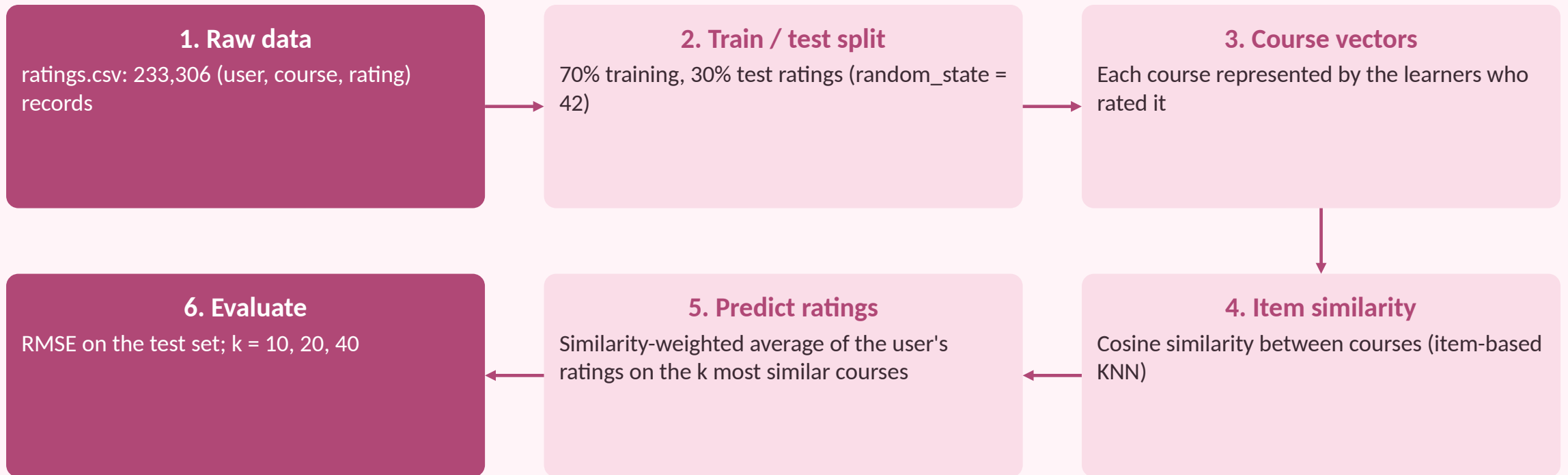
Top-10 most frequently recommended courses

#	Course ID	Title	Users
1	DS0101EN	introduction to data science	418
2	BD0101EN	big data 101	383
3	DS0105EN	data science hands on with open source tools	349
4	DS0103EN	data science methodology	325
5	PY0101EN	python for data science	311
6	ST0101EN	statistics 101	306
7	BD0111EN	hadoop 101	242
8	ML0101ENV3	machine learning with python	224
9	CC0101EN	introduction to cloud	203
10	ML0115EN	deep learning 101	191

Collaborative-filtering Recommender System using Supervised Learning

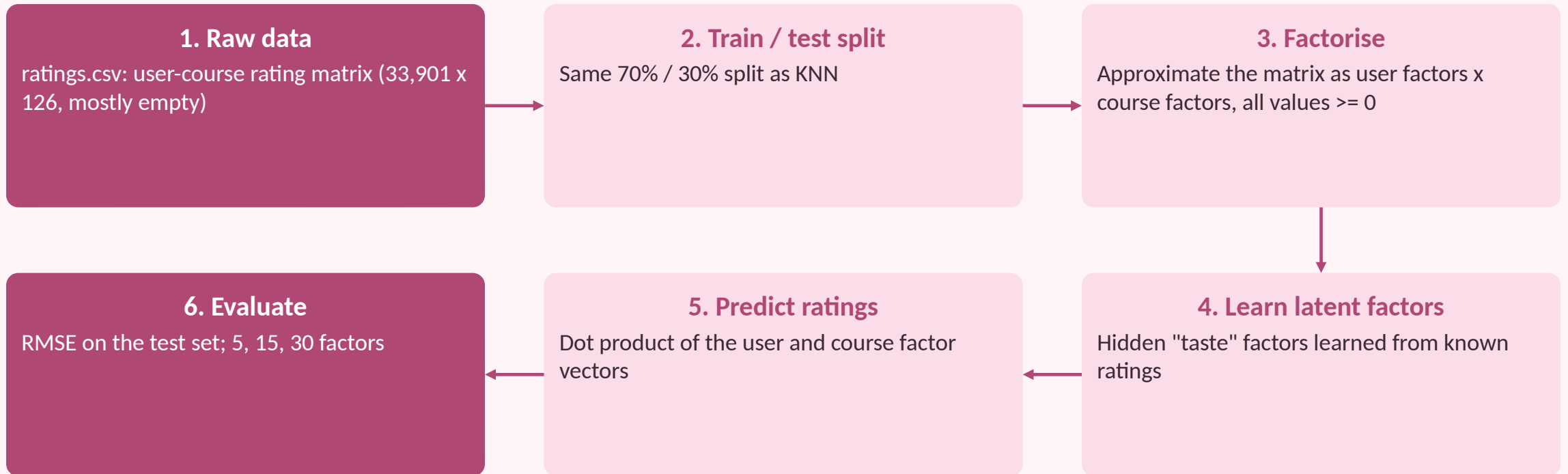


Flowchart of KNN based recommender system



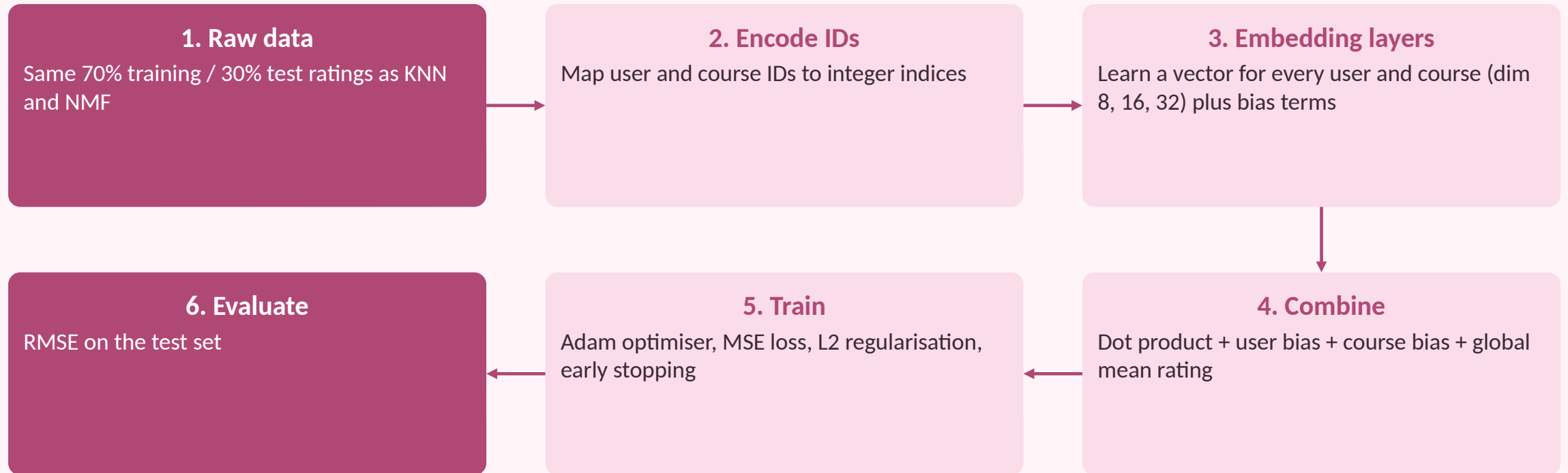
How it works: two courses are similar if the same learners enrolled in them, regardless of content. A rating is predicted from the learner's ratings of the k most similar courses. Item-based KNN was used because 126 courses are far cheaper to compare than 33,901 users. Best RMSE: 0.1933 (all k equal, since most learners rated fewer than 10 courses).

Flowchart of NMF based recommender system



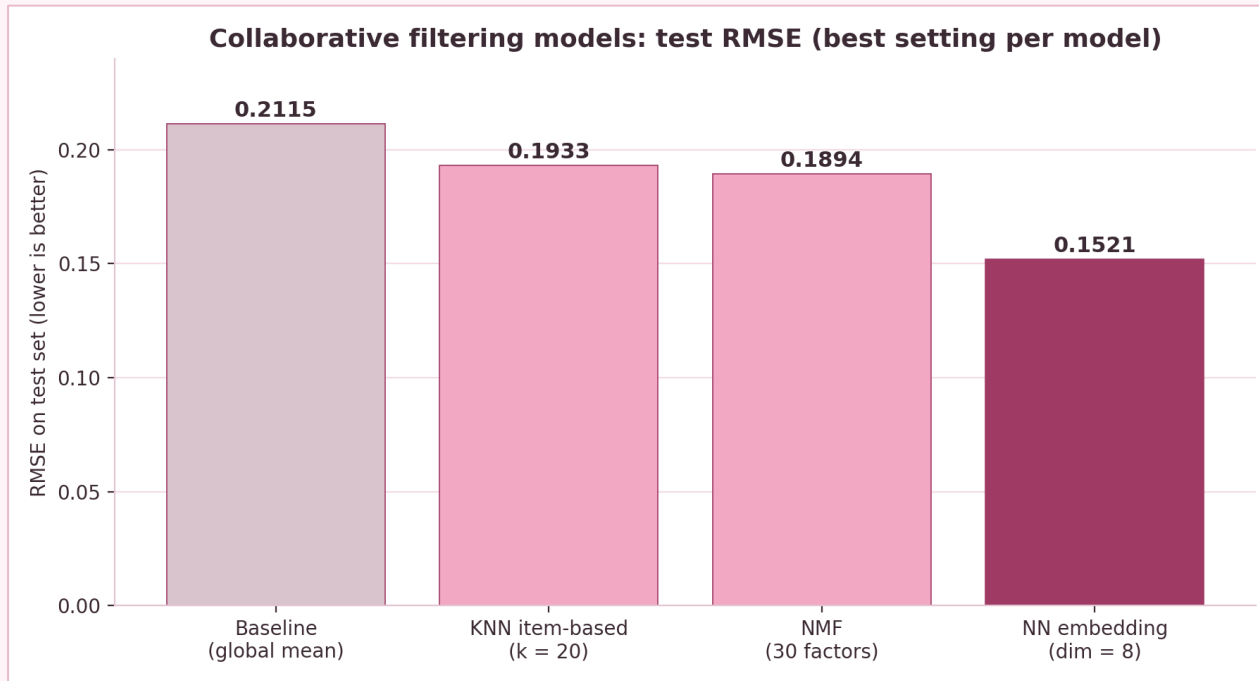
How it works: Non-negative Matrix Factorization compresses the sparse rating matrix into a small number of non-negative latent factors per learner and per course. Missing ratings are predicted by multiplying the two. Results: 5 factors underfit (RMSE 0.3614); 30 factors performed best (RMSE 0.1894).

Flowchart of Neural Network Embedding based recommender system



How it works: a Keras model learns an embedding vector for each learner and each course through backpropagation. The predicted rating is how well the two vectors match, adjusted by learner and course biases. Best result: embedding size 8, RMSE 0.1521, the lowest error of all models.

Compare the performance of collaborative-filtering models



Model	Test RMSE	vs baseline
Baseline (global mean)	0.2115	-
KNN item-based (k = 10/20/40)	0.1933	8.6%
NMF (5 factors)	0.3614	-70.9%
NMF (15 factors)	0.1986	6.1%
NMF (30 factors)	0.1894	10.4%
NN embedding (dim = 8)	0.1521	28.1%
NN embedding (dim = 16)	0.1552	26.6%
NN embedding (dim = 32)	0.1547	26.9%

- The neural embedding model (dim = 8) is the best: RMSE 0.1521, a 28.1% lower error than the baseline.
- KNN and NMF improve on the baseline by only 9-10%: 95% of ratings are 3, so there is little variance to predict.

Conclusions

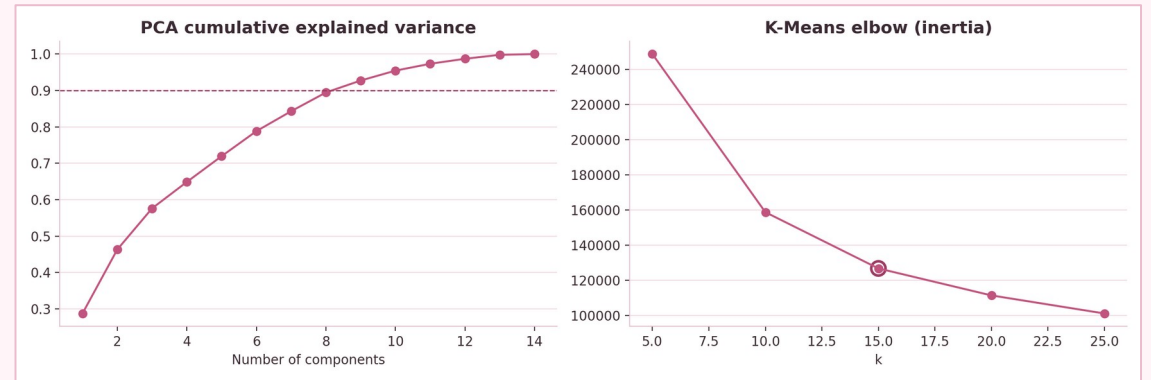
- Content-based approaches successfully generated personalized unseen-course recommendations using both learner genre profiles and course-text similarity. At the selected thresholds, the profile method recommended about 53.4 new courses per learner, while the similarity method produced about 17.3.
- Clustering provided a useful middle ground for scalable personalization. PCA reduced the user profiles to 9 components, and K-means grouped learners into 15 segments, allowing popular courses within similar learner groups to be recommended.
- For collaborative filtering, both KNN and NMF improved on the baseline RMSE of 0.2115. The best KNN achieved 0.1933, while NMF with 30 factors achieved 0.1894.
- The strongest model was the neural embedding recommender with 8 latent dimensions, achieving an RMSE of 0.1521, a 28.1% improvement over the baseline. This is the recommended model for rating prediction.
- The data also reveals important limitations: the top 20 courses account for 63.3% of all enrollments, 8,320 learners have taken only one course, ratings are overwhelmingly 3 rather than 2, and the catalogue contains several very similar or near-duplicate course titles. These create popularity bias, cold-start challenges, and limited rating variation.

Next steps: test recommendations using real learner clicks and enrollments, add more behavioral signals, improve cold-start recommendations for new learners and courses, remove or consolidate near-duplicate courses, and evaluate the system with ranking metrics such as Precision@K, Recall@K and NDCG, not RMSE alone.

Appendix

Assets

- Notebook: capstone_recommender.ipynb (Google Colab): EDA, three content-based recommenders, KNN, NMF and neural embedding models.
- Libraries: pandas, scikit-learn (CountVectorizer, PCA, K-Means), Surprise (KNNBasic, NMF), TensorFlow / Keras.
- Data: IBM Skills Network course_genre.csv, ratings.csv, user_profile.csv, rs_content_test.csv, course_processed.csv.
- Evaluation: 1,000 test users for content-based methods; 70/30 train/test split (random_state = 42) for collaborative filtering.



PCA cumulative explained variance (9 components reach 90%) and K-Means elbow curve used to choose $k = 15$.